



Transfer learning for COVID-19 predictive modeling: A multicenter study of 12 hospitals

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ABSTRACT

Purpose: To use transfer learning to predict Intensive Care Unit admission (ICU) for COVID-19 patients in a multicenter setting.

Methods: We trained XGBoost algorithms using demographic and laboratory predictors recorded during hospital admissions across 12 hospitals. Our goal was to identify the best-performing hospital and evaluate its generalization capacity through external validation in the other 11 hospitals. Additionally, this best-performing hospital served as the source domain for transfer learning and was fine-tuned by incorporating new decision trees trained with local data from each of the remaining hospitals.

Results: We observed considerable variation in ICU admission outcomes across the 12 hospitals, with local predictive performance (AUC – area under the receiver operating characteristic curve) ranging from 0.6239 to 0.9410. The best-performing model's generalization capacity, evaluated through external validation, resulted in an AUC below 0.7 in 6 hospitals. By applying transfer learning and fine-tuning this best-performing model, we found that its performance stabilized after incorporating 20 new trees trained with data from the target hospitals. This adjustment improved AUC in 9 out of the 11 hospitals, compared to the results from the external validation.

Conclusions: The study showed that transfer learning effectively leverages knowledge from the best-performing hospital to initialize and accelerate model training in other hospitals with lower performance or limited sample sizes, thereby enhancing overall adaptability and efficiency.

Introduction

The use of machine learning (ML) in healthcare has expanded rapidly, as these algorithms can learn patterns from data to support critical decision-making, such as predicting patient outcomes [1]. In developing countries, challenges such as limited diagnostic resources, regional disparities, and socioeconomic inequalities hinder data collection, highlighting the need for innovative solutions like transfer learning.

One potential approach is to develop predictive algorithms in locations with large data availability and resources and then investigate whether these algorithms can be generalized and adapted in other

settings without requiring major modifications. Transfer learning presents a promising solution in these cases, as it leverages knowledge acquired from a source domain to initialize and accelerate model training in a target domain [2,3]. This approach can be especially valuable in areas where data collection is costly or challenging, leading to limited sample sizes.

Transfer learning is frequently applied in image recognition and classification tasks using deep neural networks. A common approach involves leveraging a pre-trained model on a large dataset of images to identify general features, such as lines, corners, and edges. This model can then be fine-tuned for the specific task of disease detection in medical images, enabling more efficient and accurate results [4–6].

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While image classification is the most common application of transfer learning, this approach can also be applied to various other contexts. For instance, a study by Lee and colleagues [7] used a regression-based model to transfer valuable information from a dominant racial group (white) to enhance the prediction of a measure related to visual function in other racial groups, such as African American and Asian populations. Hammour and colleagues [8] also used a different approach of transfer learning technique based on decision trees to improve the accessibility of sleep monitoring in older adults.

Our study aims to apply machine learning methods to predict Intensive Care Unit (ICU) admissions for COVID-19 patients within a Brazilian multicenter setting. By leveraging data from multiple hospitals across diverse regions, the study captures Brazil's large socioeconomic and resource disparities, as well as varying sample sizes. Specifically, we will evaluate each hospital's predictive performance, identify the best-performing hospital, validate its generalizability across other hospitals through external validation, and determine whether transfer learning models can further improve outcomes.

Methods

A multicenter cohort study was conducted in Brazil from March to August 2020, involving patients who tested positive for COVID-19. For the current study, we analyzed the ICU admission outcome and included adult patients (aged >18 years) with positive RT-PCR for COVID-19 from 12 hospitals that provided ICU care during the pandemic. The geographic distribution of these hospitals across Brazil includes one in the Midwest, four in the Northeast, one in the North, three in the Southeast, and three in the South, with patient sample sizes varying between 56 and 1500. After removing duplicates, the final sample comprised 5840 patients.

A detailed review of all laboratory values was conducted in collaboration with clinical specialists, using predefined biological plausibility criteria. Implausible values were corrected by imputing the maximum clinically acceptable value, based on the range of normal clinical measures [9–17]. The boundaries of plausibility are presented in [supplementary material Table S1](#).

For model building, we tested three different approaches. In the first step, we trained the machine learning algorithm XGBoost [18] separately for each hospital to assess their predictive performance and identify the best-performing model. For each hospital, the data were split into training (70 %) and testing (30 %) sets and stratified on the outcome. Numeric features were standardized using z-scores. Training data were used for hyperparameter optimization via 3-fold cross-validation combined with a randomized search strategy. This hyperparameter optimization strategy allows balancing exploration of the hyperparameter space with a controlled computational cost. The hyperparameter space considered for exploration is presented in the [supplementary material Table S2](#).

Missing values were not explicitly imputed, as XGBoost inherently manages them by learning the best direction for missing values at each split node. By avoiding imputations, we keep the real structure of the data, and the information embedded in the missing value patterns, which may be present in medical scenarios. Additionally, we aimed to analyze how the proportion of missing values affects performance variation in transfer learning.

Twenty-two variables measured at hospital admission were considered as potential predictors: age, sex, heart rate, respiratory rate, systolic blood pressure, diastolic blood pressure, mean arterial pressure, temperature, hemoglobin, platelets, hematocrit, red blood cell count, mean corpuscular hemoglobin (HCM), red cell distribution width (RDW), mean corpuscular volume (MCV), leukocytes, neutrophils, lymphocytes, basophils, eosinophils, monocytes, and C-reactive protein (CRP). The contribution of these variables to outcome prediction was assessed using SHAP (Shapley Additive exPlanations) values [19].

After identifying the hospital with the best local predictive

performance, we proceeded to the second step by examining the generalization capacity of this best-performing model through external validation in the other 11 hospitals. In the third step, this best-performing model was considered the source domain for transfer learning, and it was fine-tuned by adding new trees trained with local data from each remaining hospital. The fine-tuning in this case takes the pre-trained model, i.e. the decision trees of the best-performing model, and passes it to new data. While the original decision trees remained fixed, 50 new decision trees were trained and appended based on the patterns and information obtained in the target hospital data, by minimizing the residual errors of the new dataset after passing through the pre-trained model. For these new trees added in transfer learning, we adopted the hyperparameters default of XGBoost.

The metric used to evaluate the predictive performance of all models was the area under the receiver operating characteristic curve (AUC). For comparison purposes, AUC was calculated on the same test dataset for each hospital across all three steps. A sensitivity analysis was conducted to better understand the impact of adding new trees, one at a time, to the best-performing model trained on local data at each hospital. The AUC was calculated for each of the 50 newly added trees.

In the context of transfer learning, differences between the predictors' distributions in the source and target data can pose challenges for effective algorithmic generalization. To examine these differences, we calculated skewness, kurtosis, and the coefficient of variation for the 21 quantitative predictors, and then combined these metrics by calculating their average. This approach allowed us to conduct an exploratory comparison between the source hospital and target hospitals regarding the overall behavior of data distribution.

Finally, the Pearson correlation coefficient (r) was obtained to evaluate the correlation between variation in the AUC with transfer learning related to the external validation of the best-performing model, with sample size, proportion of outcomes, and proportion of missing values in the dataset for each hospital.

The study received ethical approval from the University of São Paulo's Institutional Review Board (IRB) under reference number 32872920.4.1001.5421, which included a waiver of consent. This approval also covered the utilization of data and collaboration with all hospitals participating in the IACOV-BR network.

Results

The total sample included 5840 patients, with 54.0 % being male, and a mean age of 57.7 years (SD = 17.9). Patients were distributed across the five regions of Brazil: 539 from the Midwest (9.2 %), 2354 from the Northeast (40.3 %), 247 from the North (4.2 %), 2005 from the Southeast (34.3 %), and 695 from the South (11.9 %). [Table 1](#) provides demographic data on age and sex, categorized by hospital. The mean age ranged from 49.62 years (SD = 17.79) in the S_B hospital to 66.97 years

Table 1

Descriptive statistics for age, sex, and ICU admission outcome categorized by hospital (SE: SouthEast; NE: NorthEast; S: South, N: North, and MW: Midwest).

Hospital	sample size	Age		Male		ICU outcome	
		mean	SD	n	%	n	%
SE_A	1500	59.31	16.00	837	55.8 %	1048	69.9 %
SE_B	449	66.16	17.04	260	57.9 %	303	67.5 %
SE_C	56	66.97	17.65	33	58.9 %	24	42.9 %
NE_A	1330	49.70	18.25	590	44.4 %	245	18.4 %
NE_B	840	62.67	17.68	463	55.1 %	182	21.7 %
NE_C	112	56.99	17.36	54	48.2 %	66	58.9 %
NE_D	72	55.94	16.15	43	59.7 %	30	41.7 %
S_A	456	60.11	17.93	264	57.9 %	137	30.0 %
S_B	148	49.62	17.79	77	52.0 %	21	14.2 %
S_C	91	60.66	16.05	57	62.6 %	46	50.5 %
MW_A	539	57.99	15.78	335	62.2 %	514	95.4 %
N_A	247	56.63	17.07	138	55.9 %	75	30.4 %

(SD = 17.65) in the SE_C hospital. Datasets differed in terms of the outcome; the proportion of ICU admissions ranged from 14.2 % at S_B hospital to 95.4 % at MW_A hospital. Distributions of all quantitative predictors are presented in the [supplementary material Figure S1](#).

[Fig. 1](#) presents the Receiver Operating Characteristic curve and the area under this curve (AUC) for predicting ICU admission in each hospital, which ranged from 0.6239 to 0.9410, with the best performance observed at NE_A hospital. Although the rate of outcomes was low (18.4 %) at this hospital, the clinical and laboratory predictors were strong indicators of ICU admissions, and the number of outcomes was sufficient to learn the rules that distinguish between patients who experienced the outcome and those who did not. Hyperparameters of the optimized models for each hospital are presented in the [supplementary material Table S3](#).

[Fig. 2](#) displays the five highest Shapley values for local models across each hospital (Shapley values for all predictors are presented in [supplementary material Table S4](#)). Certain variables consistently played an important role in predicting ICU admission across multiple hospitals. For example, C-reactive protein (CRP) was important in 11 out of 12 hospitals, while age and systolic blood pressure were in 6 hospitals. Additionally, lymphocyte count, temperature, and neutrophil count were important in 4 hospitals. These findings suggest a moderate level of consistency in the variables most frequently used for outcome prediction across different hospitals

The generalization capacity of this best-performing model (NE_A) was assessed through external validation in all other hospitals. Additionally, this model was used as the source domain for transfer learning models applied to the other hospitals. [Table 2](#) presents the predictive performance of the external validation of the best-performing model (NE_A) across other hospitals, as well as the results after applying transfer learning. In this approach, the best model was used as the source domain and then updated by adding 50 new trees trained with data from each target hospital. Results indicated that the best-performing model yielded AUC metrics below 0.7 in external validation for 6 hospitals. By comparing the external validation with the transfer learning models, we observed an increase in AUC for 9 out of 11 hospitals, suggesting enhancements to the original model (NE_A), however, for NE_C, and MW_A, the AUC remained below 0.7. In two hospitals, the predictive performance declined. For hospital S_B this reduction had no impact since both external validation and transfer model showed excellent predictive performances, 0.9359 and 0.9231, respectively. However, at NE_D, the AUC decreased notably from 0.8205 to 0.7436.

The percentage change in the AUC from external validation of the best-performing model (NE_A) to the transfer learning was also calculated ([Table 2](#)). For cases where transfer learning had a positive impact, this variation ranged from 2.0 % to 28.3 %. The NE_D hospital, where transfer learning led to a decline in predictive performance (-9.4 %), is in the same region as the source model (NE_A). This hospital had a small

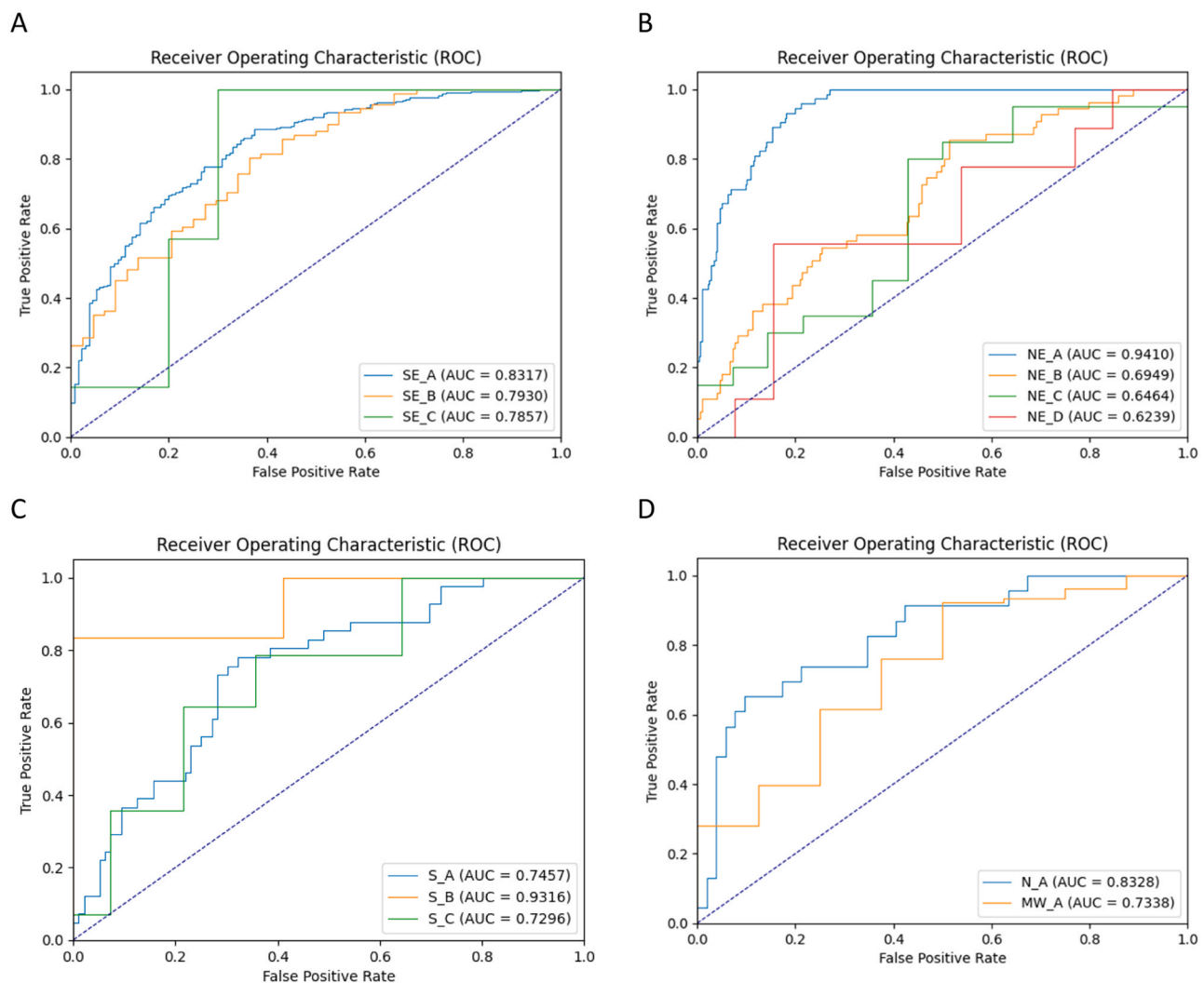


Fig. 1. The Receiver Operating Characteristic curve and the area under the curve (AUC) of the XGBoost optimized for each hospital. A: SouthEast hospitals; B: NorthEast hospitals; C: South hospitals and D: North and MidWest hospitals.

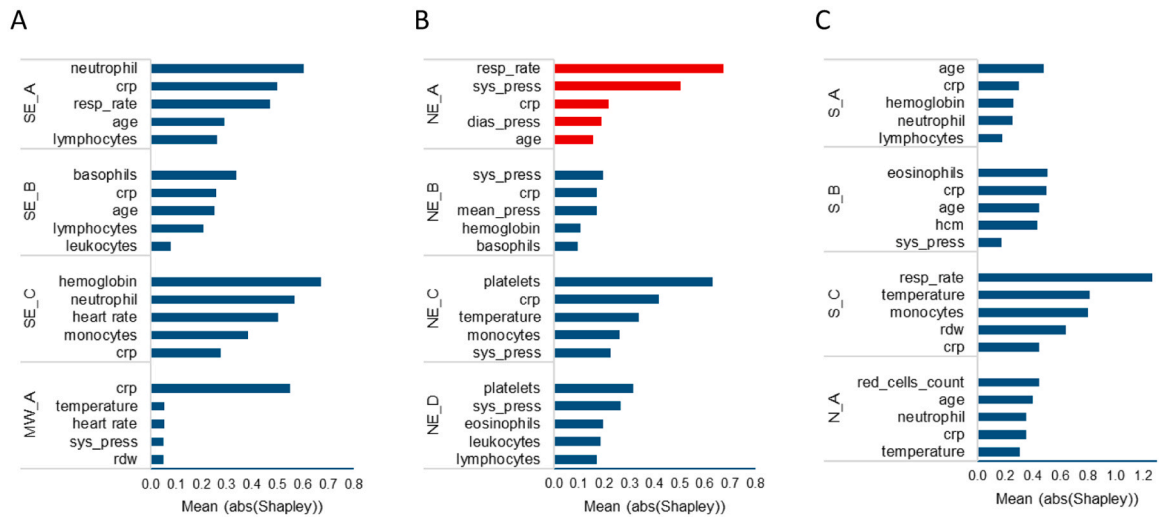


Fig. 2. Mean Absolute Shapley Values: For each hospital, the graph displays the top 5 variables with the highest contributions to outcome prediction. The results of the best-performing hospital (NE_A) are highlighted in red. A: SouthEast and MidWest hospitals; B: NorthEast hospitals; C: South and North hospitals (resp_rate: respiratory rate; crp: C-reactive protein; sys_press: systolic blood pressure; dias_press: diastolic blood pressure; mean_press: mean arterial pressure, and red_cell_counts: red cell distribution width).

Table 2

The area under the receiver operating characteristic curve (AUC) for external validation of the best model (NE_A) and transfer learning (model from NE_A updated with 50 new trees in each target hospital), variation between the two AUC measures, proportion of outcomes, missing values and combined metrics of quantitative predictors (skewness, kurtosis and coefficient of variation-cv) for each hospital.

Hospital	sample size	AUC external validation	AUC transfer learning	Variation in AUC	% outcomes	% missing	Mean skewness	Meankurtosis	Meancv
SE_A	1500	0.7745	0.8057	4.0 %	69.9 %	23.2 %	0.65	3.04	0.35
SE_B	449	0.6811	0.8272	21.5 %	67.5 %	9.6 %	0.65	7.69	0.40
SE_C	56	0.6571	0.8429	28.3 %	42.9 %	14.0 %	0.65	3.75	0.57
NE_A	1330	source model	source model	source model	18.4 %	30.1 %	0.93	4.12	0.40
NE_B	840	0.6270	0.7024	12.0 %	21.7 %	15.2 %	0.62	4.36	0.46
NE_C	112	0.6036	0.6393	5.9 %	58.9 %	22.2 %	0.73	2.40	0.50
NE_D	72	0.8205	0.7436	−9.4 %	41.7 %	1.0 %	0.71	0.82	0.45
S_A	456	0.7274	0.7574	4.1 %	30.0 %	37.7 %	0.64	6.37	0.42
S_B	148	0.9359	0.9231	−1.4 %	14.2 %	1.6 %	0.95	6.65	0.38
S_C	91	0.7602	0.7755	2.0 %	50.5 %	18.8 %	0.87	3.33	0.65
MW_A	539	0.6420	0.6778	5.6 %	95.4 %	11.7 %	1.36	13.51	1.05
N_A	247	0.6363	0.8060	26.7 %	30.4 %	24.1 %	0.45	2.96	0.30

sample size, a near-balanced proportion of outcomes, and a low rate of missing values. The combined metrics of skewness, kurtosis, and the coefficient of variation for the quantitative predictors (Table 2) of the three hospitals that achieved the best results with transfer learning

(SE_B, SE_C, and N_A) are similar to those observed in the source hospital. In contrast, hospital MW_A, which had a low AUC after the transfer, displayed metrics indicating a greater degree of asymmetry, higher kurtosis, and more variation compared to the source hospital

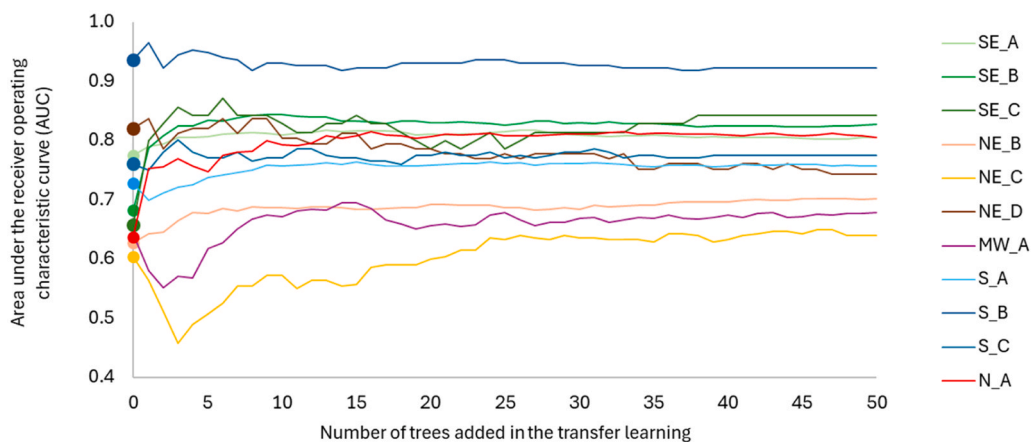


Fig. 3. The area under the receiver operating characteristic curve (AUC) for each of the 50 trees added sequentially to the best-performing model (NE_A), trained with data from each target hospital. The AUC at point 0 represents the performance of the external validation of the best-performing model in each hospital.

data. For hospital NE_D, where the increment of trees yields an important decline in AUC, we found that the predictors had much lower kurtosis in general. Interestingly, hospital NE_C also achieved a low AUC after the transfer, but its metrics were quite similar to those of the source hospital. This similarity makes it difficult to link the poor outcomes in NE_C to differences in the distributions of the predictors between the source and this hospital. Overall, the extent of this variation in AUC after transfer learning was not correlated with the hospital sample size ($r = -0.09$), proportion of outcome ($r = -0.04$), or proportion of missing data ($r = 0.10$) in the target hospital.

Fig. 3 shows that, for all target hospitals, the AUC exhibited greater variability during the addition of the first 20 new trees, after which the values stabilized.

Discussion

We applied machine learning algorithms with boosted trees to predict ICU admissions for COVID-19 patients across 12 hospitals in Brazil. The performance varied significantly, with AUC scores ranging from below 0.7 to an excellent 0.941 at NE_A hospital, which had the second-largest sample size ($n = 1330$). This variability highlights the heterogeneity across hospitals, driven by factors like clinical practices, resource availability, and demographic differences. The use of transfer learning is justified in this research scenario, as it is a multicenter study encompassing several hospitals nationwide that followed the same data collection protocol, relying on routinely collected clinical and laboratory predictors.

The best-performing hospital, NE_A, had a low rate of events, but the algorithm learned patterns that effectively distinguished between patients who did or did not require ICU admission. While external validation showed that this best-performing model did not generalize well in six hospitals, after applying the transfer learning and adding new trees trained with the target hospital, the performance metric AUC was above 0.7 in nine hospitals. One hospital, NE_D, experienced a decline from 0.8205 to 0.7436, which may be due to the lack of a clear pattern for the outcome, and also due to a difference between the source and target regarding the distribution of predictors. The slight decline observed at S_B was not relevant since both the external validation and transfer model maintained excellent performance, with AUCs above 0.92. The predictors in hospital MW_A also presented in general a greater degree of asymmetry, kurtosis and coefficient of variation compared to the source hospital data, which could explain the poor results obtained with the external validation and transfer learning. Poor outcomes in all models for NE_C likely reflect data inconsistencies rather than predictor quality, suggesting local characteristics that hindered model generalization. We did not find in this study a correlation between the size of the variation in AUC after applying transfer learning and the sample size, the proportion of outcomes, or the proportion of missing values of the target hospital.

The study demonstrated that transfer learning models require alignment between the source and target datasets to avoid negative transfer, where performance deteriorates. NE_A, the source model, performed well due to its high data quality and sample size. However, in some cases, incorporating new trees trained with local data was essential to capture local patterns and achieve satisfactory performance, and the sensitivity analysis indicated that AUC improvements plateaued after 20 trees. It is important to notice that for four hospitals, local models outperformed transfer models, reaffirming the importance of adapting to hospital-specific patterns.

Recent studies have addressed the generalizability of models trained in one scenario for predicting events in another. Rockenschaub and colleagues [20] investigated how effectively deep learning models can be transferred from one hospital to another to predict adverse events in the ICU. They observed a decline in performance when the source model was implemented at new hospitals, indicating that variations between hospitals and healthcare systems can affect a model's generalizability.

On the other hand, some studies reported positive outcomes by employing transfer learning and integrating new information from the target data. For instance, Zhang and colleagues [21] found that transfer learning improved model performance in image classification, although they noted some cases where a model trained from scratch performed equally well. Similarly, Hammour and colleagues [8] showed that fine-tuning a pre-trained model yielded satisfactory results. They utilized a boosted decision trees algorithm trained on a large dataset of scalp EEG and then fine-tuned it on a smaller dataset of ear EEG, which enhanced sleep stage prediction using the smaller dataset. While our study employed a model-based transfer learning approach, another study focused on diagnosing Alzheimer's disease used an instance-based transfer learning method, to leverage the knowledge from an auxiliary dataset to enhance classification in a target dataset that lacked sufficient training data. The findings from this study also demonstrated satisfactory performance of the transfer learning model [22].

Our study has limitations that need to be addressed. First, there was an unequal representation of different regions of the country, with two regions having only one hospital each. Additionally, the number of patients was disproportionately larger in the southeast and northeast regions. Another limitation is that clinical practices may vary between hospitals, which can affect data collection procedures, making standardization more difficult. Finally, some hospitals had small sample sizes, which posed challenges when applying machine learning methods. Our study found that the best-performing model was obtained using data from the second-largest hospital. Generally, we expect that the more examples an algorithm is trained on, the better it can learn the rules that enhance its performance. Thus, transfer learning is often anticipated to start from a source domain that has a great amount of data to help improve predictions in a target domain that has lower performance or a limited sample size. Nevertheless, a hospital with limited data can still achieve optimal results if it has clear and distinct outcomes. In such scenarios, the rules learned can be considered the source model, since our ultimate goal is to achieve the best possible outcomes.

This study, conducted across diverse Brazilian regions, indicates the need for flexible approaches in predictive healthcare models. The results confirm that no single method consistently outperforms others in every context, highlighting the value of testing multiple strategies on real-world data. For some hospitals, starting training from the ground up and utilizing a local model may be the most effective strategy. However, if this approach does not yield satisfactory predictions, the algorithmic generalization technique could be an alternative to enhance predictive performance and assist in decision-making for future patients. By leveraging the knowledge gained from the hospital that presented excellent performance, we can potentially accelerate decision-making in other hospitals. In the face of new health emergencies, this technique can be integrated with other approaches to expedite both clinical and logistical decisions, such as bed planning and management. Future research could explore ensemble models to enhance predictions in settings with high data variability.

Author Contributions

Conceptualization: ADPCF; Data curation: CS, AHAC, FBF; Formal analysis: all authors; Funding acquisition: ADPCF; Investigation: all authors; Methodology: all authors; Project administration: ADPCF; Data collection: RMW; Resources: ADPCF; Software: all authors; Supervision: ADPCF; Validation: all authors; Visualization: all authors; Roles/Writing - original draft: CS; and Writing - review & editing: all authors.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Code availability

All the code written to develop the models can be found on https://github.com/labdaps/transfer_iacovbr

Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.annepidem.2025.05.013](https://doi.org/10.1016/j.annepidem.2025.05.013).

Data availability

The data comes from 12 distinct hospitals, and it is not publicly available as it contains information of patients in accordance with the Brazilian data protection law but are available on reasonable request from the Laboratory of Big Data and Predictive Analytics in Healthcare

at the School of Public Health at the University of São Paulo (labdap-s@usp.br). The Brazilian General Data Protection Law (LGPD) restricts the public release of deidentified records to protect individuals' privacy.

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